Weekly Progress: Validation, Resampling, and Decision-Level Explanation

June 3, 2026

Executive summary. This week I closed the loop from implementation validation to empirical variability and decision-level explanation. First, the Step1c SPO+ code was validated against PyEPO references on both shortest-path and KEP oracles. Second, subset-resampling boxplots summarize the distribution of 2stage and SPO+ behavior under fixed graph pools. Third, decision-level analysis on Step2b d8 suggests that 2stage can remain competitive because many large edge-level errors are not decision-critical, while SPO+ helps when errors fall on decision-changing edges.

1 Step 1: Implementation Validation with PyEPO Reference

1.1 Motivation

The goal of this step is to check whether my implementation contains subtle bugs. The logic is simple: if my code cannot reproduce PyEPO behavior on a standard benchmark or on the same KEP oracle, then there may be a bug in the loss, gradient, optimizer update, solver interface, or metric calculation.

1.2 Shortest-Path Benchmark

I first used PyEPO's fixed-data shortest-path setup as an external benchmark. For the two-stage LR baseline, my implementation fully matches PyEPO across all fixed-data shortest-path settings:

Table 1: Shortest-path PyEPO-vs-my comparison.

Quantity	PyEPO LR vs my LR	PyEPO SPO+ vs my SPO+
Result files	24 / 24	24 / 24
Rows	240 / 240	240 / 240
Passing settings	24 / 24	19 / 24
True SPO global max diff	3.12×10^{-19}	3.61×10^{-4}
Unambiguous SPO global max diff	2.07×10^{-19}	3.61×10^{-4}
MSE global max diff	1.09×10^{-47}	2.70×10^{-3}
Epoch diff	0	0

For SPO+, the loss, gradient, and one-step update match PyEPO when the oracle solution is shared. The remaining full-CSV differences are explained by shortest-path tie-breaking: the same cost vector can have multiple optimal paths, and PyEPO's `optDataset` can store a different optimal solution vector across repeated solves even when the optimal objective is identical. Since

SPO+ gradients depend on the solution vector and not only on the objective value, this can create training divergence without implying an algebraic bug.

1.3 KEP Reference Validation

I then moved the validation to the actual KEP oracle. This is more directly relevant to Step1c because KEP is a reward-maximization problem with instance-specific graph structure. The KEP validation uses the same `CachedHybridKepModel` / Gurobi oracle and compares PyEPO-style SPOPlus reference computations with my Step1c reward-max SPO+ implementation.

Table 2: KEP PyEPO-reference vs Step1c validation summary.

Validation phase	Key comparison	Result
Phase 0 preflight	reward-max / cost-min sign adapter	all max diffs = 0
Phase 1 small setting	SPO+ forward loss	max diff = 2.9998×10^{-6}
Phase 1 small setting	SPO+ gradient and one-step SGD update	max diff = 0
Phase 2 full trajectory	$\theta_1, \theta_2, \|\theta\|$	max diff = 0
Phase 2 full trajectory	train SPO+ loss	max diff = 3.2975×10^{-6}
Phase 2 full trajectory	validation SPO+ loss	max diff = 2.2324×10^{-6}
Phase 2 full trajectory	train / validation decision gap	max diff = 0

1.4 Conclusion from Step 1

The implementation check supports the conclusion that the Step1c SPO+ algebra, reward-max sign convention, gradient scaling, optimizer update, KEP oracle interface, and evaluation path are consistent with an independent PyEPO-based reference implementation. The remaining shortest-path SPO+ CSV differences are attributable to oracle tie-breaking among multiple shortest paths, not to an identified error in the Step1c SPO+ formula.

2 Step 2: Subset-Resampling Variability

2.1 Motivation

The second step is to turn a single-seed observation into distributional evidence. The current resampling experiment intentionally isolates one source of randomness: the training subset. The graph pool, label regime, validation set, heldout set, and training initialization are fixed. Only the subset seed used to sample 50 training graphs is varied.

Thus, this experiment should be interpreted as:

> *Train-subset resampling on a fixed graph pool and fixed label regime.*

It does not yet test graph-generation seed, label-noise seed, or training initialization seed.

2.2 Experimental Setup

The first-pass resampling setting is:

Item	Setting
Train size	50
Randomness varied	subset seed only
Graph pool	fixed
Training seed	fixed
Evaluation	heldout400
Main methods	2stage selected by validation MSE; SPO+ selected by validation SPO+ loss
Primary metric	mean normalized decision gap

2.3 Generated Plots

The current plotting script summarizes the heldout400 distribution with four main outputs:

- overall gap boxplot;
- by-block gap boxplot;
- paired reduction boxplot;
- relative reduction by block.

2.4 Interpretation

These plots answer whether the SPO+ improvement is stable under different sampled training subsets from the same fixed graph pool. The paired reduction plot is especially important because it compares 2stage and SPO+ on the same subset seed, reducing confounding from subset difficulty.

The correct conclusion is limited but useful:

> SPO+ improvement appears stable under train-subset resampling on the fixed Step2 graph pool, especially in high-degree misspecification regimes.

The current experiment should not be described as full robustness over all randomness. A stronger next-stage robustness test would vary graph-generation seed, label-noise seed, or training initialization seed.

3 Step 3: Decision-Level Explanation

3.1 Research Question and Hypothesis

After completing the implementation validation and subset-resampling experiments, the next question is no longer whether the code is correct, but why the observed decision behavior occurs. In particular, the advisor's hypothesis was:

> *The two-stage model may remain competitive because bad edge-level predictions do not necessarily affect the final KEP solution.*

Therefore, the goal of this section is to move from aggregate test gaps to decision-level evidence. The main questions are:

1. When 2stage and SPO+ have similar decision quality, are they selecting the same KEP solution?

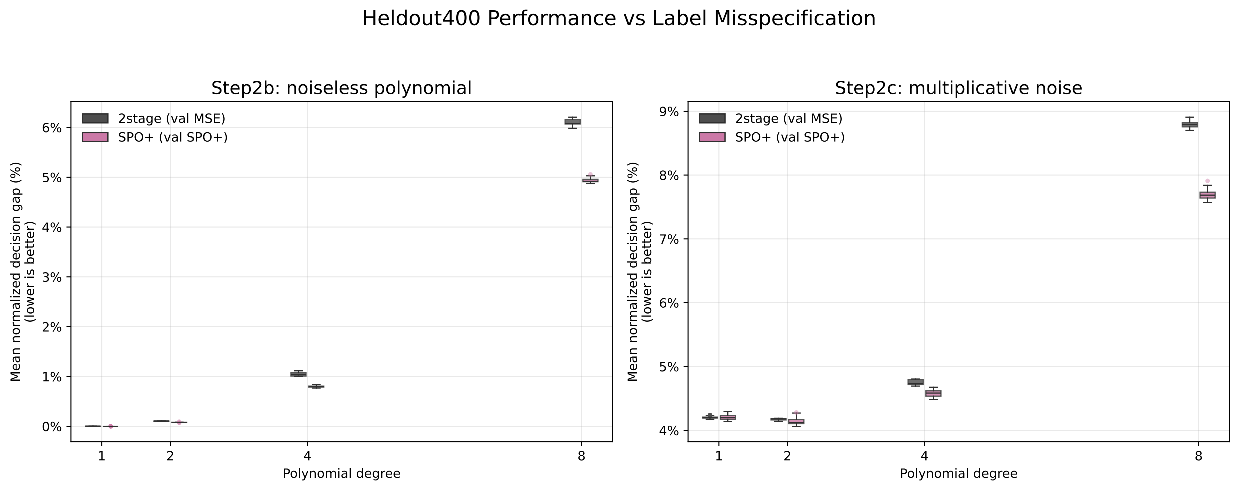

Figure 1: Heldout400 normalized decision gap by block under subset resampling.

How to read. Each box summarizes heldout400 normalized decision gaps across subset seeds for one regime/block and method. Lower boxes indicate better downstream decision quality; wider or higher boxes indicate stronger subset sensitivity.

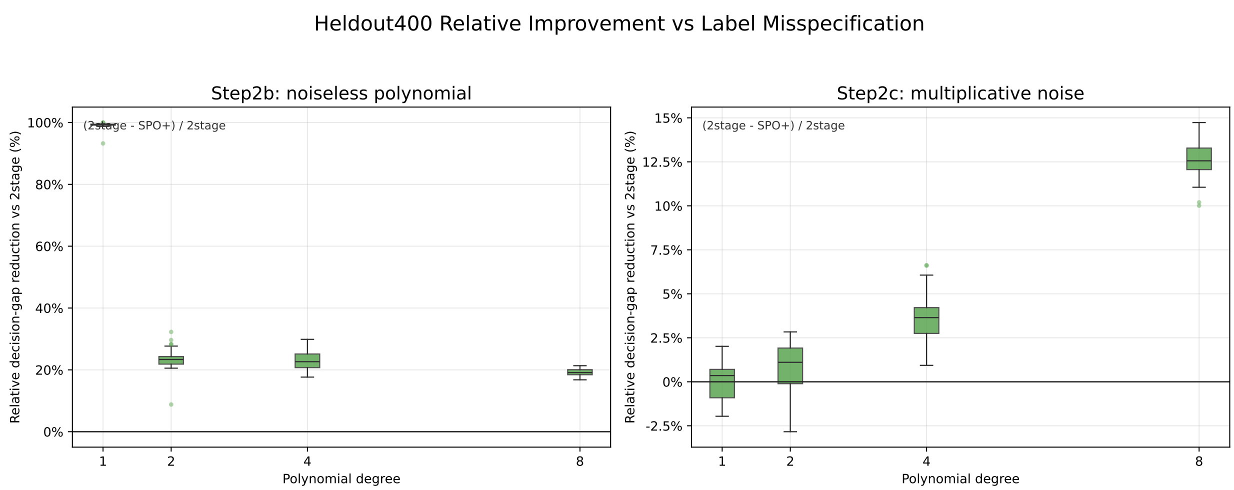

Figure 2: Heldout400 relative reduction versus 2stage by block.

How to read. Positive values mean the decision-focused method reduces normalized gap relative to the 2stage baseline on the same subset-seed protocol. Read larger positive boxes as stronger paired improvement, and values near zero as methods behaving similarly.

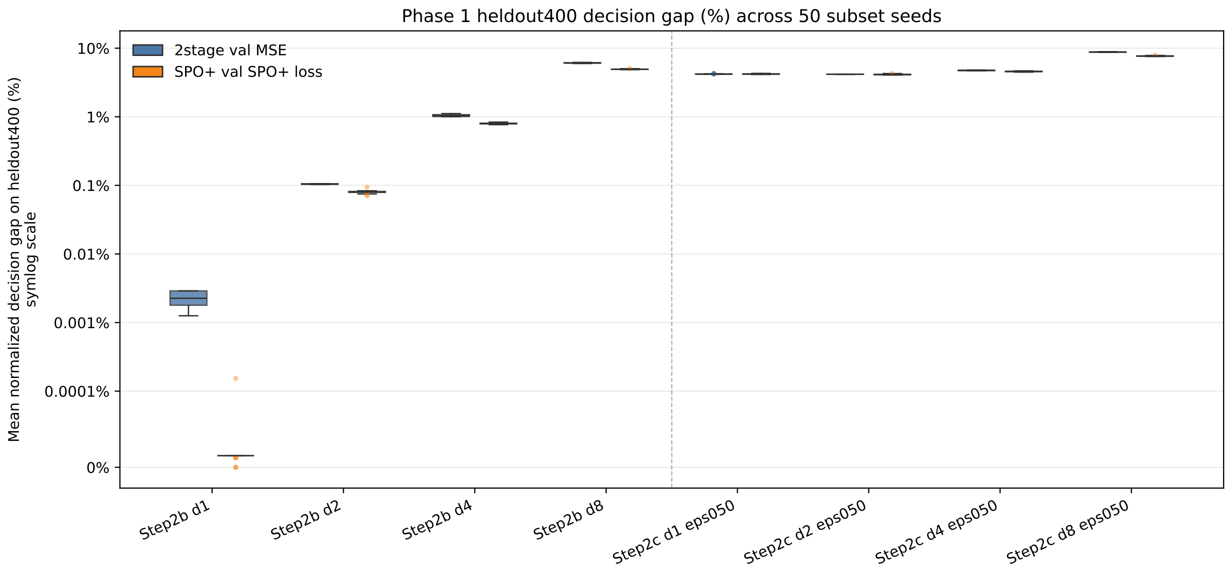

Figure 3: Overall heldout400 normalized decision gap distribution.

How to read. This plot aggregates the heldout400 gap distribution across the selected subset-resampling settings. Compare method medians and tails: lower medians indicate better typical quality, while shorter upper tails indicate fewer hard cases.

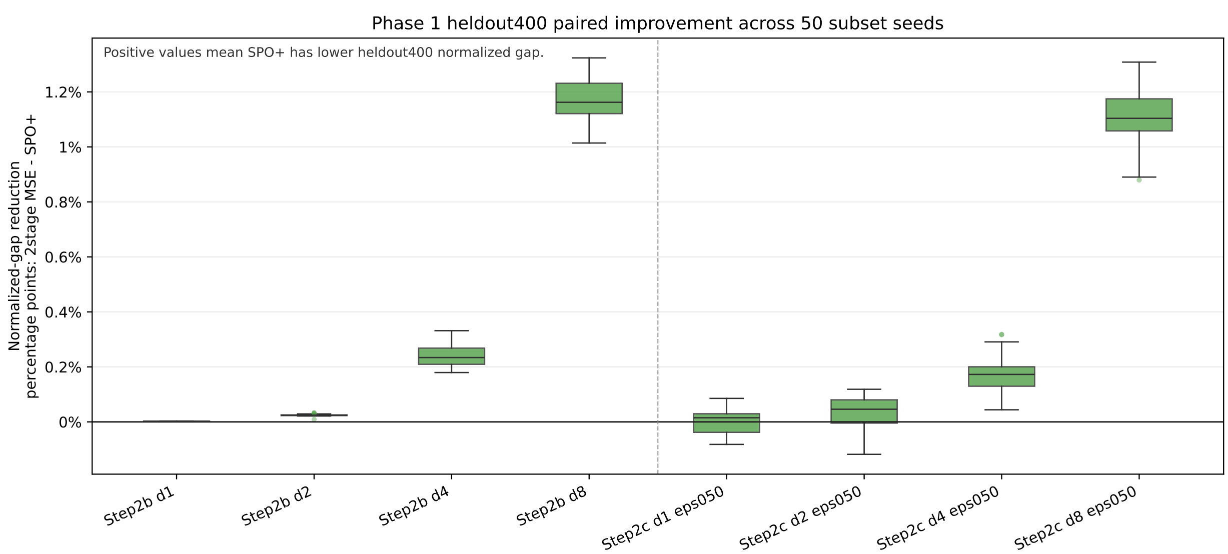

Figure 4: Paired heldout400 improvement: 2stage gap minus SPO+ gap.

How to read. Each value compares 2stage and SPO+ on the same subset seed. Positive values mean SPO+ has a lower normalized decision gap; the paired view reduces confounding from subset difficulty.

2. If they select different solutions, are these solutions near-optimal alternatives under the true objective?
3. Are large edge-level prediction errors actually decision-critical, or do they often occur on edges that do not affect the selected solution?
4. When SPO+ improves over 2stage, does it reduce errors on decision-changing edges?

The central distinction is between *raw prediction error* and *decision-critical prediction error*. A large edge prediction error matters only if it changes the selected KEP solution or appears in the symmetric difference between the predicted and oracle solutions.

3.2 Analysis Scope

This first-pass analysis focuses on the most difficult noiseless polynomial misspecification setting:

$$\texttt{Step2b d8}, \qquad n_{\text{train}} = 50, \qquad \texttt{evaluation} = \texttt{heldout400}.$$

The two compared methods are:

$$\text{2stage selected by validation MSE} \qquad \text{vs.} \qquad \text{SPO+ selected by validation SPO+ loss.}$$

This is intentionally a narrow setting. The goal is not to claim a universal mechanism across all regimes, but to obtain a clean first-pass explanation in a high-misspecification setting where 2stage and SPO+ differences are meaningful.

3.3 Analysis Design

The decision-level analysis has four stages:

1. **Seed selection.** Select representative subset seeds from the resampling results: large SPO+ improvement, weak/borderline improvement, and easy low-gap cases.
2. **Per-graph decision replay.** For each selected seed and each heldout graph, replay the 2stage and SPO+ checkpoints and record the selected KEP solutions.
3. **Edge-level criticality analysis.** For each graph, compute edge-level prediction errors and classify each edge according to whether it appears in the oracle solution, predicted solution, or their symmetric difference.
4. **Mechanism extraction.** Use summary plots, correlation analysis, near-tie analysis, and case studies to identify why 2stage sometimes remains competitive and when SPO+ helps.

The analysis produced the following files:

Output	Rows / files
selected case seeds	9 rows
per-graph decision comparison	7,200 rows
edge-level criticality table	879,642 rows
graph-level criticality summary	7,200 rows
case-study index	9 rows
case-study edge tables	9 tables
margin / near-tie analysis	3,600 rows
observed candidate ranking	10,800 rows
decision-critical MSE correlations	12 rows

3.4 Seed Selection: Why These 9 Seeds

The selected seeds were chosen from the subset-resampling results, not manually from individual graphs. The idea is to cover three qualitatively different situations:

- **Large improvement:** 2stage has relatively high decision gap, while SPO+ substantially improves over it.
- **Weak or borderline improvement:** SPO+ does not strongly improve over 2stage.
- **Easy low-gap:** both 2stage and SPO+ have low decision gaps.

The selected seeds are:

$$\begin{aligned} \text{large improvement:} &\quad \{1, 25, 22\}, \\ \text{weak/borderline improvement:} &\quad \{27, 21, 30\}, \\ \text{easy low-gap:} &\quad \{41, 16, 33\}. \end{aligned}$$

This selection makes the later case studies interpretable: we can compare cases where SPO+ clearly helps, cases where both methods are already good, and cases where the improvement is weak.

3.5 Decision Replay: From Aggregate Gap to Per-Graph Decisions

Aggregate heldout400 gaps are useful, but they do not reveal what decision was made on each graph. Therefore, for each selected seed, I replayed the trained 2stage and SPO+ checkpoints on every heldout graph and recorded:

- the oracle solution $y^\star$ under the true reward w_{true};
- the 2stage-selected solution y_{2stage};
- the SPO+-selected solution $y_{\text{SPO+}}$;
- true objective values and decision gaps;
- solution overlap metrics such as Jaccard similarity with the oracle solution.

For a method m, the decision gap is:

$$\text{gap}_m = w_{\text{true}}^\top y^\star - w_{\text{true}}^\top y_m,$$

and the normalized gap is:

$$\text{normalized gap}_m = \frac{w_{\text{true}}^\top y^\star - w_{\text{true}}^\top y_m}{\left|w_{\text{true}}^\top y^\star\right| + \epsilon}.$$

To ensure that this replay follows the same evaluation path as the original Step2 heldout400 metrics, the replayed per-graph gaps were checked against each seed's existing `metrics/test_per_graph.csv`. The maximum differences were:

$$\max|\Delta\text{gap}| = 5.07 \times 10^{-5}, \qquad \max|\Delta\text{normalized gap}| = 1.65 \times 10^{-7}.$$

These differences are within the intended tolerance, so the following analysis is based on the same decision path as the existing Step2 evaluation.

3.6 Edge Criticality: From Raw Error to Decision-Changing Error

The advisor's hypothesis is about edge-level prediction errors. For each edge e, I computed:

$$\mathrm{err}_e^{(m)} = \left|\hat{w}_e^{(m)} - w_{\mathrm{true},e}\right|,$$

where m is either 2stage or SPO+. Each edge is then classified using the following binary indicators:

Indicator	Meaning
`in_opt`	edge appears in the oracle solution $y^\star$
`in_2stage`	edge appears in y_{2stage}
`in_spoplus`	edge appears in $y_{\mathrm{SPO+}}$
`in_2stage_symdiff`	edge appears in $y^\star \triangle y_{\mathrm{2stage}}$
`in_spoplus_symdiff`	edge appears in $y^\star \triangle y_{\mathrm{SPO+}}$

The symmetric difference is the key object:

$$y^\star \triangle y_m = \{e : y_e^\star \neq y_{m,e}\}.$$

Edges in this set are precisely the edges where the predicted solution differs from the oracle solution. Thus, prediction errors on these edges are more likely to be decision-critical than prediction errors on edges that are never selected.

3.7 First-Pass Aggregate Findings

Across the 9 selected seeds and 400 heldout graphs, the first-pass summary is:

Table 3: Decision-level summary on selected Step2b d8 heldout400 cases.

Metric	2stage	SPO+
Mean normalized decision gap	0.06087	**0.04938**
Mean Jaccard with oracle solution	0.60580	**0.63655**
Exact oracle-solution rate	17.81%	**23.58%**
Top-10-error edges selected rate	22.87%	**19.05%**
Top-10-error edges in symmetric difference	9.39%	**7.02%**
Observed 2stage/SPO+ solution relationship		
Same 2stage/SPO+ solution rate	55.08%	
Mean Jaccard between 2stage and SPO+	0.82707	
Median Jaccard between 2stage and SPO+	1.00000	
Any-method near-tie rate, threshold = 0.01	9.17%	
Correlation with normalized gap, Pearson / Spearman		
All-edge MSE	0.095 / 0.023	-0.021 / -0.075
MSE on symmetric-difference edges	0.390 / 0.528	0.272 / 0.372
Top-10-error symmetric-difference rate	**0.570 / 0.667**	**0.557 / 0.639**

There are three main observations.

First, SPO+ has a lower mean normalized decision gap and higher average solution overlap with the oracle solution. This confirms that SPO+ is not only improving a numerical loss but is also selecting edge sets that are closer to the oracle solution.

Second, the top-10-error symmetric-difference rate is relatively low. For 2stage, only 9.39% of top-10-error edges enter the symmetric difference with the oracle solution. For SPO+, this rate is 7.02%. This suggests that many large edge-level prediction errors are not decision-changing.

Third, all-edge MSE is weakly correlated with decision gap. In contrast, error measures restricted to symmetric-difference edges are much more predictive. This suggests that raw prediction accuracy is not enough to explain decision quality; what matters is whether errors occur on decision-changing edges.

3.8 Summary Plots

The next three figures show the first-pass decision-level diagnostics. They are kept as separate full-width figures so each diagnostic can be read on its own.

The MSE-vs-gap plot shows that SPO+ can have higher edge-level MSE but lower decision gap. The solution-overlap plot shows that lower overlap often increases gap, but there are also medium-overlap points with small gap, suggesting near-optimal alternative solutions. The high-error criticality plot shows that the highest-error edges are selected or enter the symmetric difference only at limited rates.

3.9 Case Studies: Three Mechanisms

To make the aggregate patterns interpretable, I selected three types of graph case studies.

Case A: Bad Prediction but Irrelevant

Case A: Bad prediction but irrelevant.

Seed	Graph	2stage gap	All-edge MSE	Top-10-error symdiff rate
41	`G-234.json`	0	19.6328	0.0
41	`G-994.json`	0	18.1776	0.0
25	`G-1516.json`	0	17.7713	0.0

These graphs have large all-edge prediction error under the 2stage model, but the selected solution is still optimal. The top-error edges do not enter the optimal-vs-predicted symmetric difference. Therefore, the errors are large in prediction space but irrelevant in decision space.

Figure 8: Large prediction errors can be non-critical.

How to read. These rows have high all-edge MSE but zero 2stage decision gap. The key column is the top-10-error symmetric-difference rate: it is zero, so the largest prediction errors do not affect the decision-changing edge set.

This case directly supports the hypothesis that bad edge estimates do not necessarily affect the final solution.

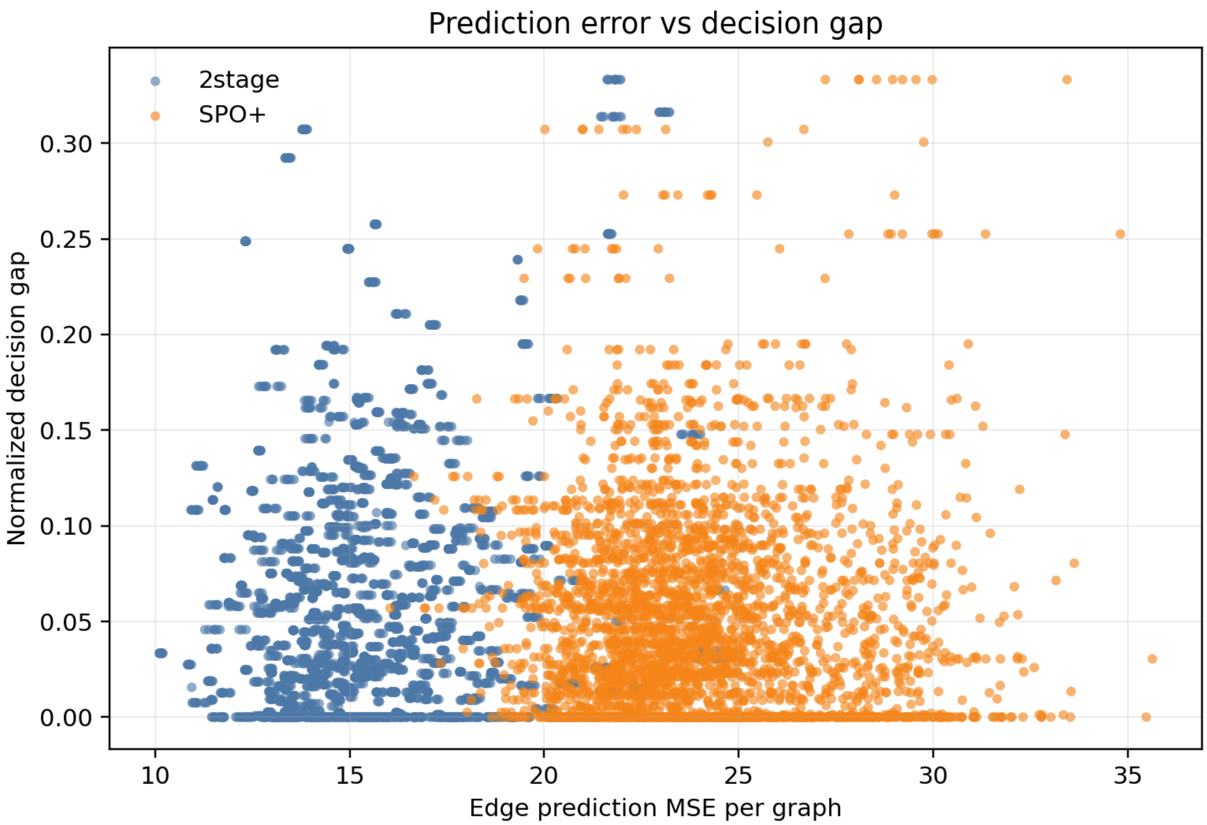

Figure 5: Prediction MSE versus normalized decision gap.

How to read. Each point is a method/graph evaluation from the selected Step2b d8 heldout400 cases. If raw prediction MSE fully explained decision quality, points would align strongly upward; the weak pattern shows that edge error alone is not enough to explain KEP regret.

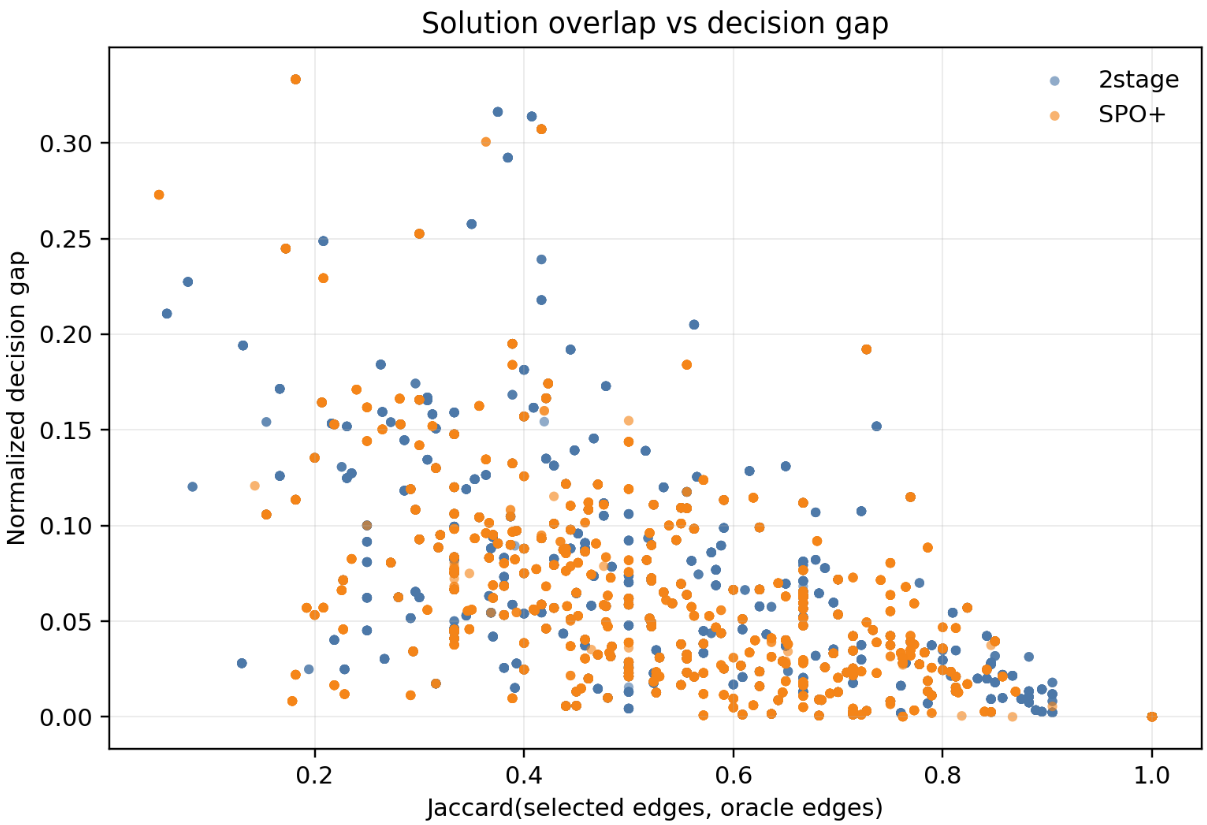

Figure 6: Solution overlap with the oracle versus normalized decision gap.

How to read. The x-axis measures edge-set overlap with the oracle solution, and the y-axis measures normalized gap. Lower-overlap points with small gaps indicate near-optimal alternative exchanges: the selected solution can differ while remaining close in true objective.

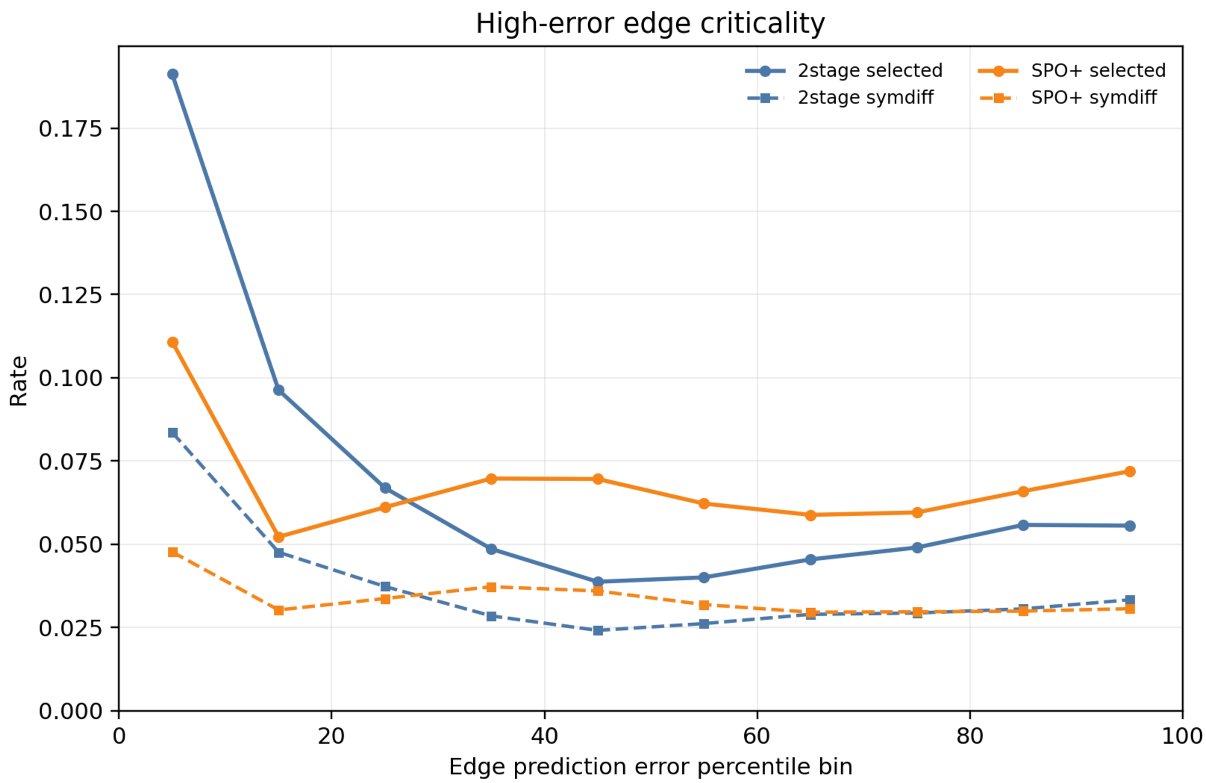

Figure 7: High-error edge selected and symmetric-difference rates.

How to read. The leftmost error-rank bins contain the largest prediction errors. Compare selected-rate and symmetric-difference-rate curves: high-error edges matter most when they enter the selected solution or the predicted-vs-oracle symmetric difference.

Case B: Different Solution but Near-Optimal

Case B: Different solution but near-optimal.

Seed	Graph	2stage gap	Jaccard with oracle	Same as oracle?
1	`G-696.json`	0.000644	0.6818	No
1	`G-1372.json`	0.001174	0.6087	No
1	`G-607.json`	0.001175	0.7143	No

Here, the 2stage solution is not identical to the oracle solution, and the Jaccard overlap is only moderate. However, the true objective gap is nearly zero. This suggests that the graph contains near-optimal alternative exchanges.

Figure 9: Different selected solutions can be near-optimal alternatives.

How to read. These rows are not exact oracle solutions and have only medium Jaccard overlap, but their normalized decision gaps are close to zero. This supports the near-tie story: solution identity can change while true objective remains almost unchanged.

This explains another way in which 2stage can remain competitive: it may select a different exchange, but that exchange is nearly equivalent under the true objective.

Case C: SPO+ Fixes Decision-Critical Errors

Case C: SPO+ fixes 2stage decision-critical errors.

Seed	Graph	2stage gap	SPO+ gap	2stage top-10 symdiff	SPO+ top-10 symdiff
1	`G-392.json`	0.313849	0	0.3	0.0
30	`G-1560.json`	0.292305	0.011964	0.3	0.0
1	`G-39.json`	0.316226	0.080459	0.2	0.1

In these graphs, the 2stage model makes errors that affect decision-changing edges. SPO+ substantially reduces the decision gap and also reduces the involvement of high-error edges in the symmetric difference.

Figure 10: SPO+ helps when prediction errors are decision-critical.

How to read. Compare the 2stage and SPO+ gap columns together with the top-10-error symmetric-difference rates. SPO+ reduces the gap precisely in cases where 2stage's large errors enter decision-changing edges.

This case explains when SPO+ is useful: it helps most when prediction errors fall on the edges that change the selected KEP solution.

3.10 Near-Tie and Margin Analysis

The case studies suggest that some different selected solutions may be near-optimal alternatives. To quantify this, I compared the observed candidate solutions:

$$y^\star, \qquad y_{\text{2stage}}, \qquad y_{\text{SPO+}}.$$

This is not an enumeration of all feasible KEP solutions; it only ranks the solutions actually observed in the replay. With a near-tie threshold of 0.01 in normalized gap, the first-pass summary is:

Quantity	Value
Paired graph rows	3,600
Observed candidate rows	10,800
Mean $\lvert \text{obj}_{2stage} - \text{obj}_{SPO+} \rvert$	4.3366
Median $\lvert \text{obj}_{2stage} - \text{obj}_{SPO+} \rvert$	0
Mean Jaccard($2stage, SPO+$)	0.8271
Median Jaccard($2stage, SPO+$)	1.0000
Same 2stage/SPO+ solution rate	55.08%
2stage different-solution near-tie rate	6.08%
SPO+ different-solution near-tie rate	6.58%
Any-method near-tie rate	9.17%

This supports the near-optimal-alternative explanation. In more than half of the observed graph-seed pairs, 2stage and SPO+ select the same solution. When they differ, a nontrivial subset of cases still has normalized gap below 1%, meaning that different solution identities can correspond to very similar true objectives.

3.11 Decision-Critical MSE Correlations

Finally, I tested whether different prediction-error summaries explain normalized decision gap. The main comparison is between all-edge MSE and decision-focused error summaries such as MSE on symmetric-difference edges.

Table 4: Correlation between prediction-error summaries and normalized decision gap.

Predictor	2stage Pearson / Spearman	SPO+ Pearson / Spearman
All-edge MSE	0.095 / 0.023	-0.021 / -0.075
MSE on selected edges	-0.260 / -0.237	-0.112 / -0.066
MSE on symmetric-difference edges	0.390 / 0.528	0.272 / 0.372
Top-10-error symmetric-difference rate	**0.570 / 0.667**	**0.557 / 0.639**

The strongest predictor for both methods is the rate at which top-error edges enter the symmetric difference. This provides the most direct evidence for the decision-critical-error hypothesis: prediction error matters primarily when it changes the selected solution.

3.12 Answer to the Advisor's Hypothesis and Limitations

The first-pass evidence supports the hypothesis. The reason 2stage can remain competitive is not that it predicts every edge accurately. Instead, many large edge-level errors are non-critical because they occur on edges that are not selected or do not enter the symmetric difference. In addition, KEP graphs may contain near-optimal alternative solutions, so two different selected edge sets can have very similar true objectives.

SPO+ appears most helpful when the 2stage prediction errors fall on decision-changing edges. In those cases, SPO+ reduces both the decision gap and the involvement of high-error edges in the symmetric difference.

This is still a first-pass analysis. The current limitations are:

- only `Step2b d8` is analyzed;
- only train size 50 and heldout400 are used;

- only 9 representative subset seeds are selected;
- only 2stage and SPO+ are compared;
- FY is not yet included;
- `Step2c` multiplicative-noise regimes are not yet analyzed;
- the near-tie analysis uses only observed candidates $y^\star$, $y_{2\text{stage}}$, and $y_{\text{SPO+}}$;
- adversarial SPO+ solutions y_{adv} are not yet included.

The next step is to extend this same analysis to `Step2c d8 eps050`, add FY, and include y_{adv} to inspect how SPO+ gradients push the predictions away from bad decision-changing solutions.

4 Overall Conclusions and Next Steps

4.1 Conclusions

This week's work forms a closed loop:

1. **Code validation.** The Step1c SPO+ implementation aligns with PyEPO reference computations on both shortest-path and KEP oracles. The remaining shortest-path full-training SPO+ discrepancy is explained by oracle tie-breaking among multiple shortest paths.
2. **Distributional evidence.** The subset-resampling box plots show how 2stage and SPO+ vary across different 50-graph training subsets on a fixed graph pool.
3. **Mechanistic explanation.** The decision-level analysis suggests why 2stage can remain competitive: many prediction errors are not decision-critical, and some different selected exchanges are near-optimal. SPO+ improves most when errors occur on decision-changing edges.

Table 5: Safe claims supported by the current evidence.

Claim	Evidence	Limitation
Step1c SPO+ implementation is reliable.	PyEPO-reference validation on shortest-path and KEP; KEP Phase 2 trajectory equivalence.	Shortest-path full CSV SPO+ is affected by oracle tie-breaking.
SPO+ improves over 2stage in high-degree subset-resampling settings.	Heldout400 boxplots and paired reductions on fixed graph pools.	Only subset seed varied; not graph seed or label seed.
2stage remains competitive because many errors are non-critical.	Case A/B/C, high-error symdiff rates, and decision-critical MSE correlations.	First-pass evidence on selected Step2b d8 seeds only.

4.2 Limitations

The resampling analysis currently varies only the training subset seed. It does not yet vary graph-generation seed, label-noise seed, or training initialization seed. The decision-level analysis is currently limited to `Step2b d8`, train size 50, heldout400, selected representative seeds, and 2stage vs SPO+. It does not yet include FY, `Step2c d8 eps050`, full graph-seed variation, or adversarial SPO+ solutions y_{adv}.

4.3 Next Steps

The next report should extend the explanation in three directions:

1. Apply the same decision-level pipeline to `Step2c d8 eps050` to check whether the same non-critical-error mechanism holds under multiplicative label noise.
2. Add FY as a third method and compare whether FY-selected solutions are closer to SPO+, 2stage, or the oracle solution.
3. Add y_{adv} analysis to inspect which edges the SPO+ gradient pushes away from or toward.